VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY, NAGPUR

Department of Computer Science and Engineering

M.Tech. (Computer Science & Engineering – Generative AI)

MINI PROJECT SYNOPSIS

Title

Development of a Retrieval-Augmented Generation Based Academic Question Answering System Using Large Language Models

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Submitted By

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Semester: II

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1. Introduction

The increasing availability of digital educational resources, lecture notes, research papers, and technical reports has created a need for intelligent systems that can efficiently retrieve and present information. Traditional keyword-based search systems often fail to understand the semantic meaning of user queries and therefore provide less relevant results.

Recent advances in Generative Artificial Intelligence and Large Language Models (LLMs) have enabled systems that can retrieve relevant information and generate context-aware responses. Retrieval-Augmented Generation (RAG) combines information retrieval techniques with generative models, reducing hallucinations and improving answer quality.

This project proposes the development of an Academic Question Answering System based on Retrieval-Augmented Generation that allows users to upload academic PDF documents and obtain accurate answers through natural language interaction.

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2. Problem Statement

Students and researchers spend considerable time searching through lengthy academic documents to find relevant information. Existing search systems rely primarily on keyword matching and often fail to provide direct answers.

There is a need for an intelligent document understanding system capable of retrieving relevant information from academic documents and generating accurate, source-supported responses.

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3. Objectives

The objectives of the proposed work are:

1. To extract textual information from academic PDF documents.
2. To generate semantic embeddings for document understanding.
3. To build a vector database for efficient information retrieval.
4. To implement a Retrieval-Augmented Generation framework.
5. To integrate Large Language Models for answer generation.
6. To provide source citations for generated answers.
7. To develop a user-friendly web interface.

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4. Literature Survey

Paper 1

Lewis et al. (2020), Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

The authors proposed Retrieval-Augmented Generation (RAG), combining neural retrieval with sequence generation to improve factual accuracy and knowledge-intensive tasks.

Paper 2

Devlin et al. (2018), BERT: Pre-training of Deep Bidirectional Transformers

BERT introduced contextual language representations that significantly improved various natural language understanding tasks.

Paper 3

Touvron et al. (2023), LLaMA: Open and Efficient Foundation Language Models

LLaMA demonstrated that compact open-source language models can achieve strong performance while remaining deployable on limited hardware.

Research Gap

Most existing systems either provide keyword-based search or generic chatbot functionality. Few systems offer domain-specific academic document understanding with retrieval-supported answer generation.

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5. Proposed Methodology

Phase 1: Document Processing

- Upload PDF documents.
- Extract text content.
- Clean and preprocess text.

Phase 2: Knowledge Base Construction

- Divide documents into semantic chunks.
- Generate vector embeddings.
- Store embeddings using FAISS.

Phase 3: Retrieval System

- Convert user queries into embeddings.
- Retrieve top relevant chunks.

Phase 4: Answer Generation

- Provide retrieved context to an LLM.
- Generate accurate responses.
- Display source references.

Phase 5: User Interface

- Develop Streamlit-based web application.
- Support conversational interaction.

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6. System Architecture

User Query

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Streamlit Interface

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PDF Processing Module

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Text Chunking

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Embedding Generation

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FAISS Vector Database

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Retriever

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Large Language Model (Gemma/Llama)

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Answer Generation with Sources

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User Response

7. Hardware and Software Requirements

Hardware

- Intel Core i5 Processor
- 16 GB RAM
- Windows 11 Laptop

Software

- Python 3.12
- Streamlit
- FAISS
- Sentence Transformers
- Ollama
- GitHub
- Visual Studio Code

8. Expected Outcomes

The developed system will:

- Allow natural language interaction with academic documents.
- Improve information retrieval efficiency.
- Generate context-aware answers.
- Demonstrate practical applications of Generative AI.
- Serve as a foundation for future academic assistants.

9. Work Plan

Activity	Duration
Literature Survey	2 Weeks
System Design	1 Week
PDF Processing Module	2 Weeks
Embedding & Retrieval Module	2 Weeks

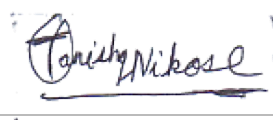
Activity	Duration
LLM Integration	2 Weeks
Testing & Validation	2 Weeks
Report Writing	2 Weeks

10. Expected Deliverables

1. Working RAG-based Academic QA System.
2. Source Code Repository on GitHub.
3. Project Report.
4. Presentation Slides.
5. Demonstration Video.

11. References

1. Lewis, P., et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, NeurIPS 2020.
2. Devlin, J., et al. BERT: Pre-training of Deep Bidirectional Transformers, NAACL 2019.
3. Touvron, H., et al. LLaMA: Open and Efficient Foundation Language Models, 2023.
4. FAISS Documentation.
5. Hugging Face Documentation.
6. LangChain Documentation.

Student Signature 

Guide Signature

Head of Department